

Candidate's Name:

Signature:

Random No.						Personal No.		

553/2

BIOLOGY

Paper 2

(practical)

July./Aug. 2026

2 $\frac{1}{2}$ hours

THE BIOLOGY MYSTERY MOCK EXAMINATION 2026

Uganda Certificate of Education

INSTRUCTIONS TO CANDIDATES:

- This paper consists of **two** examination items. Respond to **both** items in the spaces provided.
- Drawings should be made in the spaces provided. Use a **sharp** pencil for your drawings. Coloured pencils or crayons should **not** be used.
- **No** additional sheets of writing paper are to be inserted in this booklet.
- Work on additional sheets of paper will **not** be scored

FOR SCORERS' ONLY

Item No.	Basic code	Weighted score	Scorer's initials
1	A1		
	H1		
	V1		
	AP1		
	P1		
	D1		
	DP1		
	C1		
	IR1		
2	I2		
	S2		
	D2		
TOTAL WEIGHTED SCORE			

Item 1

Nazigo Rubber Manufacturing Company uses a specialized biological cleaning agent to remove residue-hardened latex from its processing tanks by converting it to substances that dissolve in the cleaning solution that is rinsed afterwards. Recently, the factory has been struggling with high maintenance costs because the tanks are sometimes not thoroughly cleaned.

The factory manager suspects that the cleaning solution, obtained from different sources, has fluctuating pH, which reduces the effectiveness of the biological agent added.

To optimize the process, the factory plans to conduct an investigation using two indicator solutions, **K** and **L**, which work differently as follows:

Indicator solution	Procedure of use	Expected observations of solution	
		Latex present	Latex absent
K	A small volume of the solution below 4cm ³ is mixed with 3-4 drops of K.	Blue-black	Brown/yellow
L	If a small volume of the solution below 4 cm ³ is mixed with equal amounts of L and the mixture is boiled.	Blue	Orange

During investigations, 2-4 cm³ of biological agent was to be mixed with an equal volume of cleaning agents of different pH solutions, followed by 2-4 cm³ of fresh latex extract, and the mixtures will be incubated at 35- 40 °C for a minimum of 15 minutes. After incubation, indicators **K** and **L** are used to determine the presence or absence of latex.

You are provided with a Latex extract labelled **A**, a biological agent labelled **R**, alkaline and acidified cleaning solutions.

Task: Design and carry out an investigation to determine how pH conditions affect the conversion of latex into soluble substances, and recommend a suitable cleaning solution with appropriate pH conditions for cleaning the factory's processing tanks.

Your investigation should include the aim, hypothesis, variables, materials, procedures, results, analysis and conclusion.

(a) Aim, hypothesis, variables and materials

Aim: To investigate the effect of **alkaline and acidified cleaning solutions (pH)** on colour changes of indicator solutions K and L when mixed with mixtures of latex solution A and biological agent R to recommend the suitable pH medium for cleaning the processing tanks for the factory.

Or

To investigate the effect of pH on the colour changes of indicators K and L due to conversion of latex to soluble substances using biological agent R to recommend a suitable pH medium for cleaning processing tanks for the factory

Scoring criteria

A2 = 06, A1 = 05, A0 = 00

- A2 - States "the what" and "the why"
- A1 – states only "the what" or only the "why"
- A0 –States none or an aim not related to the scenario
- Reject aim if independent and dependent variables are not mentioned

Hypothesis: An alkaline solution causes the colours of indicators K and L to turn brown and orange, respectively, indicating the absence of latex due to its conversion to soluble substances, while an acidified solution causes indicators K and L to turn blue-black and blue, respectively, indicating the presence of latex.

Scoring criteria

H2 = 06, H1 = 05, H0 = 00

- H2 –Mentions alkaline and acidic solutions, compares the 2 pH media & links to a comparative outcome with mention of the word "pH"/"medium"
- H1 – Does not mention one solution/ doesn't compare or misses the word pH/medium
- H0 –Does not mention any of the above.
- Reject aim if it is not related to the scenario or doesn't mention independent and dependent variables

Variables

Independent variables: pH of cleaning solutions

Dependent variables: Colour changes of indicators K and L

Controlled variables: Time of incubation, Temperature, Volume of indicators, Volume of biological agent and latex solution

Scoring criteria

V3 = 05, V2 = 04, V1 = 03, V0 = 00

- V3 – Any one well categorised in each of the 3 categories
- V2 – Any one well categorised in any 2 of the categories
- V1 – Any one well categorised in any category/list of uncategorized /interchanged/any correct Variable placed in a wrong category.
- V0 – Wrong variables/not related to scenario

Materials

Test tubes, beakers, solution A, Solution R, alkaline solution, acidified solution, indicators K and L, thermometer, water, heat source, stop clock/watch.

Scoring criteria

- To be scored in the procedure
- Only essential materials needed
- Score implied use of apparatus

Procedure of the investigation**Incubation of the mixture**

- Two test tubes were labelled (i) and (ii)
- To test tube (i), 2 cm³ of biological agent R was added, followed by 2 cm³ of alkaline cleaning solution.
- To test tube (ii), 2 cm³ of biological agent R was added, followed by 2 cm³ of acidified solution.
- To each test tube (i) and (ii), 4 cm³ of latex solution was added
- A water bath was prepared using a thermometer, hot water, cold water, and beakers at 35⁰C to 40⁰C.
- Test tubes (i) and (ii) were transferred into a water bath and immediately started a stop clock/watch, and the setup was left for 15 minutes.
- After 15 minutes, tests on the mixtures in test tubes (i) and (ii) were carried out using indicators K and L.
- 1cm³ of the mixture was measured using a measuring cylinder into a clean test tube, followed by 3 drops of indicator solution K
- To 1 cm³ of latex solution in a clean test tube, 1 cm³ of indicator solution L was added, and the mixture was boiled.
- Colour changes of indicators K and L were immediately recorded

Scoring criteria

P = 06 (5cp, 4cv, 4cm)

P = 04 (2-4cp, 3-2cv, 3-2cm)

P = 03 (1cp, 1cv, 1cm, any cp/cm/cv, all wrong cp with some cv/cm)

P = 00 (No cp/cv/cm)

- Stop awarding Cp if coherence is lost (R plus ph medium plus A)
- Do not award cv if units are wrong
- Award cm for only the 5 essential materials
- Award cp for implied use of a material.

The 7 CPs of the procedure are as follows;

- ✓ Use of test tube (when labelled or when a solution is added to it)
- ✓ Measurement of biological agent/solution R
- ✓ Measurement of pH media/solutions alkaline and acidic medium
- ✓ Measurement of substrate/latex solution A
- ✓ Measure the temperature of the water bath or use a thermometer
- ✓ Use of a stopwatch to measure time
- ✓ Final tests using the indicators K and L correctly

(b) Results, analysis and conclusion. (Interpretation of results, IR)**Results**

Test tube	Observations after 15 minutes		Deductions
	indicator K	indicator L	
(i)	<u>Brown</u>	<u>Orange</u>	<u>Latex absent</u>
(ii)	<u>Blue-black</u>	<u>Blue/green</u>	<u>Latex absent</u>

Scoring criteria

D2= 04, D1 = 03, D0 = 00

DP1= 01, DP0=00

- D2 - for a correct trend of results,
- D1- any 2 correct/wrong trend/inaccurate data
- D0 – wrong data/not related to experiment
- DP1 - data is precisely presented in a table
- DP0- data not presented in a table
- Reject if latex is absent in test tube with acidified solution
- Deductions are part of data analysis

Data analysis and conclusion

- Mixture from Test tube (i) lacks latex while that from test tube (ii) contains latex.

- An Alkaline solution provides most **suitable or favourable pH** for conversion of latex to soluble substances because it **activates biological agent R**.
- An acidic solution **inhibits the action of biological agent R** by **denaturing it**, thus providing an **unfavourable pH** for its action
- Therefore, conversion of latex to soluble substances during cleaning of processing tanks is **affected by changes in pH**, with **alkaline pH** being the **most favourable** to the activity of biological substance.

Recommendation: The company should use alkaline cleaning solutions only.

(c) Explanation and advice to the rubber manufacturing company.

- The company should always **carry out pH tests on cleaning solutions** before use.
- If cleaning solutions are acidic, the company may **use buffers or neutralise the acid using alkaline chemicals**

Scoring criteria

IR3= 05, IR2 = 04, IR1 = 03, IR0 = 00

- IR3 = 4IR points plus a recommendation
- IR2 = 2-3 IR points plus a recommendation and advice/ 3IR points without a recommendation
- IR1 = 1 IR point plus a recommendation or advice /1-2 IR points without a recommendation/only Recommendation without IR points
- IR0 = 0 No IR point and no recommendation or no advice.
- Deny recommendation if more than 1 solutions are suggested.
- Award IR points on the deduction even if presented in results table
- Award IR points on nature of pH and effect of pH of the cleaning solutions
- Nature and effect of the pH should be comparative to depict the trend.
- Score IR independent of data except where test tubes are labelled with different letters

Item 2

Okello is a porter at the Busia border who spends his day carrying heavy sacks of produce on his back. Over time, he has developed chronic lower back and neck pain. His doctor at the local hospital explained that the weight-bearing part of his spine is under extreme tension and might be damaged. Articular surfaces and structures for muscle attachment and protection of the spinal cord on the vertebral column are believed to have been damaged.

You are provided with specimens X, Y and Z which were obtained from the vertebral column for examination.

Task:

- (a) Using features of the specimens X, Y and Z, identify the regions from which each specimen was obtained.

Specimen	Body region	Reason
X(cervical)	➤ <u>Neck region</u>	➤ <u>Vertebroarterial canals in the transverse processes</u> ➤ <u>Divided transverse processes</u> ➤ <u>Short neural spine</u>
Y(thoracic)	➤ <u>Thoracic region</u>	➤ <u>Extra facets (capitular and tubercular facets)</u> ➤ <u>Long and curved neural spine</u>
Z(lumbar)	➤ <u>Abdominal region</u>	➤ <u>Long transverse processes</u> ➤ <u>Thick centrum</u>

Scoring criteria

I3 = 06, I2 = 04, I1 = 02, I0 = 00

- I3 = Correct identity plus any 2 correct reasons in each case
- I2 = correct identity plus 1 correct reason in each case or 2 correct reasons in only two identities
- I1 = any two or one correct identities or reasons
- I0 = No identity/wrong identity with wrong reasons

- (b) (i) Explain how damage to the structures of specimen Z and X might affect their roles in the regions they were obtained and lead to Okello's back pain.

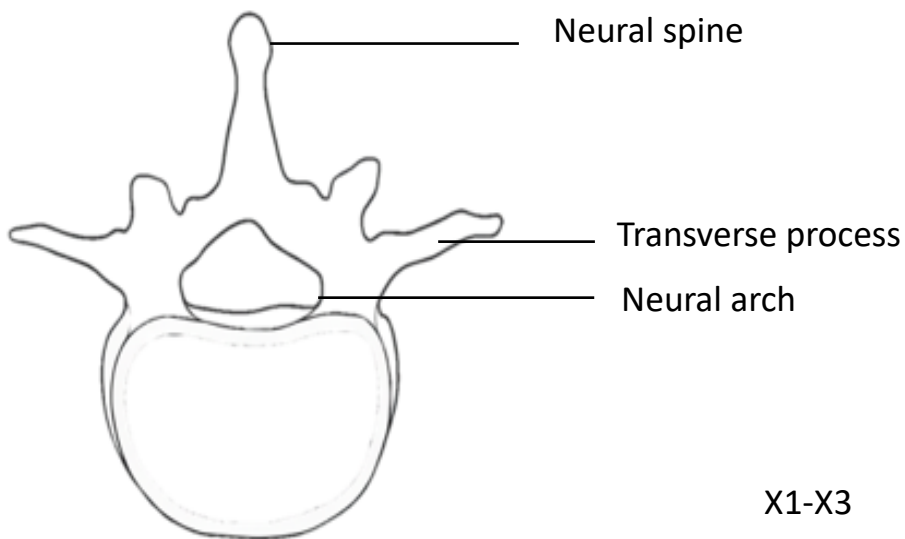
- Damage to the neural arch **exposes the spinal cord and spinal nerves to direct mechanical damage/pressure, leading to pain**
 - Damage of pre-and post zygapophysis would make these **surfaces rough to increase friction** during articulation of vertebral bones.
 - Damage to transverse processes **reduces surfaces for muscle attachment, leading to pain whenever a person tries to bend**
- (ii) Explain how the unique features on specimen Y that are absent on specimens X and Z would affect the body of the organism if damaged.
- **Damage to costal facets/extra facets increases friction at surfaces where ribs articulate with thoracic vertebrae, making breathing difficult.**

S2= 05, S1 = 02, S0 = 00

- S2 = any two disrupted suitabilities showing structure & disrupted function of structure in (i) plus one disrupted function in (ii)
- S1 = any one disrupted suitability showing structure & disrupted function of structure in (i) or one disrupted function in (ii)
- S0 = wrong suitability with no structure or function stated

(c) In the space provided below, make a drawing of the specimen that was obtained from the lower back and label structures for muscle attachment and those for protection of the spinal cord.

Drawing showing anterior view of specimen Z to show parts for muscle attachments and protection of the spinal cord



Scoring criteria: Drawing skills

T – Title indicating structures intended to show (for muscle attachment and protection of the spinal cord)

O – Correct overall shape and no broken lines

M – Should have X or Y; range from X1-X4

A – The lengths and sizes of parts should be proportional (longest transverse process, large centrum, neural spine is half the length of the transverse process, curved transverse process)

N – No rubbing marks, no stray lines, no shading, no dirty marks on paper

Labelling – any 2 intended structures; Deny the 2 labelling scores if any other structure is labelled.

D4=06, D3= 05, D2= 04, D1= 03, D0= 00

- D4 –Any 4 drawing attributes with 2 correct labels
- D3 –Any 3 drawing attributes with 2 correct labels or 4 drawing attributes without labels

- D2 –Any 2 drawing attributes with 2 correct labels or 3 drawing attributes without labels
- D1 –Any 1 drawing attribute with/without a label or 2 drawing attributes without a label
- D0 –No drawing attributes and no label
- Reject structures which are not specialised for muscle attachment and spinal cord protection
- If separate parts for locomotion are drawn, score only the title.

CONFIDENTIAL INFORMATION

- K- Iodine solution
- L- Benedict's solution
- A- 1% starch solution
- R- 2% amylase solution
- 1 M HCl solution
- 1 M NaOH
- Thermometer
- Two beakers
- Heat source
- Y- cervical vertebra
- X- thoracic vertebra
- Z- lumbar vertebra